

Robotics

AIG240NAA.05262.2264

Course Faculty

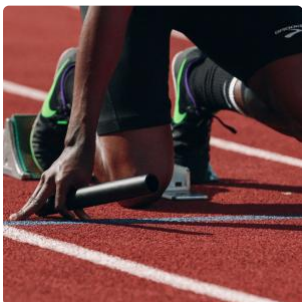


Details & Actions



-  **Roster** (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/outline/roster)
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-  **Course Description** (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/outline/description)
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-  **Attendance** (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/outline/attendanceGrade?courseId=_779956_1)
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-  **Books & Tools** (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/outline/booksAndTools?parentId=_25939574_1)
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Course Content (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/outline/content-search)



Resources to Help You Get Started

Welcome to your course! This module contains helpful course information and resources



Land Acknowledgment: Honouring the Land

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939584_1?view=content&state=view)

-  Welcome Message and Course Orientation
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939585_1?view=content&state=view)
-  Faculty Contact Information
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939587_1?view=content&state=view)
-  Curriculum Integration of Priority Areas
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939588_1?view=content&state=view)
-  Help Navigating Learn@Seneca and Technical Requirements
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939589_1?view=content&state=view)
-  Alternatives for Accessing Content in Learn@Seneca
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939590_1?view=content&state=view)
-  Student Resources, Help, and Supports
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939591_1?view=content&state=view)
-  Academic Integrity and Copyright at Seneca Polytechnic
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939592_1?view=content&state=view)
-  Generative AI
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_25939593_1?view=content&state=view)
-  Need Help / Give Help! ...
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/discussion/_25939579_1?view=discussions)

 **AIG 240 - Robotics | Summer 2026** (<https://syl.senecapolytechnic.ca/syllabi/2026-summer/aig240-naa-5262/index.html>)

 **Teams Access & Online Classes**
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/document/_26464332_1?view=content&state=view)

 **Lectures**



-  Chapter 1 - Introduction to Robotics
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26424790_1?courseId=_779956_1)
  ...
-  Chapter 2 - Robotics Systems and Design.pdf
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26463872_1?courseId=_779956_1)
  ...
-  Chapter 3 - Robot Operating System (ROS).pdf
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26488770_1?courseId=_779956_1)
  ...
-  Chapter 4 - Levels of Autonomy in Robotics.pdf
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_2648878_1?courseId=_779956_1)
  ...
-  Chapter 5 - Locomotion and Motion Control.pdf
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26488790_1?courseId=_779956_1)
  ...
-  Chapter 6 - Robot Perception and Sensors.pdf
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26488792_1?courseId=_779956_1)
  ...
-  Chapter 7 - Autonomous Navigation and Path Planning.pdf
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26488793_1?courseId=_779956_1)
  ...
-  Chapter 8 - Artificial Intelligence in Robotics.pdf
 (https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26488794_1?courseId=_779956_1)
  ...

Lectures Recordings

-  Chapter 1 - Introduction to Robotics
 (https://teams.microsoft.com/l/meetingrecap?driveld=b%21b7w6VhwheU-iRbPyWrYqITTsFFsyKoFGiEGNDG2r80WPQH4HoxjRLtuO1WQ3iDM&drivelteld=01T2PEDFM5BYTWWS75ZJEI IWCK3B4QVNC7&sitePath=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+1+-+Introduction+to+Robotics-20260505_144819-Meeting+Recording.mp4%3Fweb%3D1&fileUrl=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+1+-+Introduction+to+Robotics-20260505_144819-Meeting+Recording.mp4%3Fweb%3D1&threadId=19%3A5ab7685006d04b47a1)

347058b098564e%40thread.tacv2&organizerId=000d1dd8-2f57-472f-8357-b5e284c75cb6&tenantId=eb34f74a-58e7-4a8b-9e59-433e4c412757&callId=38e840c1-dd7b-4b7b-8088-c9906162ca16&threadType=topic&meetingType=Unknown&channelType=Standard&replyChainId=1778006843822&subType=RecapSharingLink_RecapChiclet)

↔ Chapter 2 - Robotics and Systems and Design

(https://teams.microsoft.com/l/meetingrecap?driveId=b%21b7w6VhwheU-iRbPyWrYqITTsFFsyKoFGiEGNDG2r80WPQHM4HoxjRLtuO1WQ3iDM&driveItemId=01T2PEDFOWKFNMKQIFYZD344ZPWXIUUMUM&sitePath=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+2+-+Robotics+Systems+and+Design-20260512_143127-Meeting+Recording.mp4%3Fweb%3D1&fileUrl=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+2+-+Robotics+Systems+and+Design-20260512_143127-Meeting+Recording.mp4%3Fweb%3D1&threadId=19%3A5ab7685006d04b47a1347058b098564e%40thread.tacv2&organizerId=000d1dd8-2f57-472f-8357-b5e284c75cb6&tenantId=eb34f74a-58e7-4a8b-9e59-433e4c412757&callId=cdfdb8f5-af1e-43cd-9604-e65191656d74&threadType=topic&meetingType=Unknown&channelType=Standard&replyChainId=1778610520404&subType=RecapSharingLink_RecapChiclet)

↔ Chapter 3 - Robot Operating System (ROS)

(https://teams.microsoft.com/l/meetingrecap?driveId=b%21b7w6VhwheU-iRbPyWrYqITTsFFsyKoFGiEGNDG2r80WPQHM4HoxjRLtuO1WQ3iDM&driveItemId=01T2PEDFJ2ACYEDT3UXVAKKKYZUY33P5HH&sitePath=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+3+-+Robot+Operating+System+%28ROS%29-20260519_150958-Meeting+Recording.mp4%3Fweb%3D1&fileUrl=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+3+-+Robot+Operating+System+%28ROS%29-20260519_150958-Meeting+Recording.mp4%3Fweb%3D1&threadId=19%3A5ab7685006d04b47a1347058b098564e%40thread.tacv2&organizerId=000d1dd8-2f57-472f-8357-b5e284c75cb6&tenantId=eb34f74a-58e7-4a8b-9e59-433e4c412757&callId=469a8f2d-42e7-4df5-a64d-b9c48b378ff2&threadType=topic&meetingType=Unknown&channelType=Standard&replyChainId=1779217652188&subType=RecapSharingLink_RecapChiclet)

↔ Chapter 4 - Levels of Autonomy in Robotics

(https://teams.microsoft.com/l/meetingrecap?driveId=b%21b7w6VhwheU-iRbPyWrYqITTsFFsyKoFGiEGNDG2r80WPQHM4HoxjRLtuO1WQ3iDM&driveItemId=01T2PEDFJ2ACYEDT3UXVAKKKYZUY33P5HH&sitePath=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+4+-+Levels+of+Autonomy+in+Robotics-20260519_150958-Meeting+Recording.mp4%3Fweb%3D1&fileUrl=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+4+-+Levels+of+Autonomy+in+Robotics-20260519_150958-Meeting+Recording.mp4%3Fweb%3D1&threadId=19%3A5ab7685006d04b47a1347058b098564e%40thread.tacv2&organizerId=000d1dd8-2f57-472f-8357-b5e284c75cb6&tenantId=eb34f74a-58e7-4a8b-9e59-433e4c412757&callId=469a8f2d-42e7-4df5-a64d-b9c48b378ff2&threadType=topic&meetingType=Unknown&channelType=Standard&replyChainId=1779217652188&subType=RecapSharingLink_RecapChiclet)

d=01T2PEDFLLHTHGVXJDJBF3SOFUH7HG6ATH&sitePath=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FLevels+of+Autonomy+in+Robotics-20260602_143010-Meeting+Recording.mp4%3Fweb%3D1&fileUrl=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FLevels+of+Autonomy+in+Robotics-20260602_143010-Meeting+Recording.mp4%3Fweb%3D1&threadId=19%3A5ab7685006d04b47a1347058b098564e%40thread.tacv2&organizerId=000d1dd8-2f57-472f-8357-b5e284c75cb6&tenantId=eb34f74a-58e7-4a8b-9e59-433e4c412757&callId=63ceae04-3663-4ae9-ba5d-46044a436c8e&threadType=topic&meetingType=Unknown&channelType=Standard&replyChainId=1780424810553&subType=RecapSharingLink_RecapChiclet)

↔ Chapter 5 - Locomotion and Motion Control

(https://teams.microsoft.com/l/meetingrecap?driveId=b%21b7w6VhwheU-iRbPyWrYqITTsFFsyKoFGiEGNDG2r80WPQHM4HoxjRLtuO1WQ3iDM&driveItemId=01T2PEDFJC64RA4JKON5B2RBLSOJEEDXMW&sitePath=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+5+-+Locomotion+and+Motion+Control-20260609_142955-Meeting+Recording.mp4%3Fweb%3D1&fileUrl=https%3A%2F%2Fseneca.sharepoint.com%2Fsites%2FAIG240-RoboticsSummer2026%2FShared+Documents%2FOnline+Sessions%2FRecordings%2FView+Only%2FChapter+5+-+Locomotion+and+Motion+Control-20260609_142955-Meeting+Recording.mp4%3Fweb%3D1&threadId=19%3A5ab7685006d04b47a1347058b098564e%40thread.tacv2&organizerId=000d1dd8-2f57-472f-8357-b5e284c75cb6&tenantId=eb34f74a-58e7-4a8b-9e59-433e4c412757&callId=0deb0cf7-b175-44e5-8fb7-18ff5e2c3a1e&threadType=topic&meetingType=Unknown&channelType=Standard&replyChainId=1781029719934&subType=RecapSharingLink_RecapChiclet)

Labs



-  JetAuto-User-Manual.pdf  ...
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/file/_26452102_1?courseId=_779956_1)
-  jetauto_ws.zip (https://learn.senecapolytechnic.ca/bbcswebdav/pid-26523429-dt-content-rid-218671738_2/xid-218671738_2?xythos-download=true) ...
-  Lab1_ROS_Melodic_TurtleSim_Engineering.ipynb ...
(https://learn.senecapolytechnic.ca/bbcswebdav/pid-26452100-dt-content-rid-

217103633_2/xid-217103633_2?xythos-download=true)

 Lab1_ROS2_Jazzy_Ubuntu24_TurtleSim_Engineering.ipynb ...
(https://learn.senecapolytechnic.ca/bbcswebdav/pid-26452131-dt-content-rid-217103964_2/xid-217103964_2?xythos-download=true)

This Lab version is only for those students who have decided to install earlier Ubuntu version and ROS 2.

 Lab 2 - ROS_Graph_Nodes_Topics_Services_Parameters_TurtleSim
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26464019_1/overview?courseId=_779956_1)

Due date: 5/15/26, 11:59 PM (EDT) | Formative

 Lab 3 - ROS_Workspace_Packages_PubSub_CPP_Python
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26488773_1/overview?courseId=_779956_1)

Due date: 5/22/26, 11:59 PM (EDT) | Formative

 Lab 4 - Robot_Model_RViz_and_Gazebo
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26488812_1/overview?courseId=_779956_1)

Due date: 6/14/26, 11:59 PM (EDT) | Formative

 Lab 5 - JetAuto Robot Inspection Connection and Movement
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26488822_1/overview?courseId=_779956_1)

Due date: 6/21/26, 11:59 PM (EDT) | Formative

 Lab 6 - SLAM with JetAuto
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489342_1/overview?courseId=_779956_1)

Due date: 7/19/26, 11:59 PM (EDT) | Formative

 Lab 7 - Autonomous Navigation with move_base
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489356_1/overview?courseId=_779956_1)

Due date: 7/26/26, 11:59 PM (EDT) | Formative

Projects



 Project 1 - TurtleSim Controller
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26488778_1/overview?courseId=_779956_1)

Due date: 5/31/26, 11:59 PM (EDT)

 Project 2 - JetAuto Gazebo Square Motion
(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26488815_1/overview?courseId=_779956_1)

Due date: 6/21/26, 11:59 PM (EDT)



Project 3 - JetAuto Real World Motion Control

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489330_1/overview?courseId=_779956_1)

Due date: 7/19/26, 11:59 PM (EDT)



Project 4 - JetAuto Mapping

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489351_1/overview?courseId=_779956_1)

Due date: 7/26/26, 11:59 PM (EDT)



Project 5 - JetAuto Navigation

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489358_1/overview?courseId=_779956_1)

Due date: 8/9/26, 11:59 PM (EDT)

Quizzes



Quiz 1

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26464074_1/overview?courseId=_779956_1)

Due date: 5/15/26, 9:45 AM (EDT) | Time limit: 15 minutes

Chapter 1 - Introduction to Robotics



Quiz 2

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26488779_1/overview?courseId=_779956_1)

Due date: 5/22/26, 9:45 AM (EDT) | Time limit: 15 minutes

Chapter 2 - Robotics Systems and Design



Quiz 3

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489360_1/overview?courseId=_779956_1)

Due date: 6/5/26, 9:45 AM (EDT) | Time limit: 15 minutes

Chapter 3 - Robot Operating System (ROS)



Quiz 4

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489362_1/overview?courseId=_779956_1)

Due date: 6/12/26, 9:45 AM (EDT) | Time limit: 15 minutes

Chapter 4 - Levels of Autonomy in Robotics



Quiz 5

(https://learn.senecapolytechnic.ca/ultra/courses/_779956_1/assessment/_26489365_1/overview?courseId=_779956_1)

Due: 11/17/2025 5:00 PM (EST) | Total: 10/45 | 10/10